

15  $A, B$  and  $C$  are three points such that

$$\overrightarrow{AB} = 3\mathbf{a} + 4\mathbf{b}$$

$$\overrightarrow{AC} = 15\mathbf{a} + 20\mathbf{b}$$

(a) Prove that  $A, B$  and  $C$  lie on a straight line.

(2)

$D, E$  and  $F$  are three points on a straight line such that

$$\overrightarrow{DE} = 3\mathbf{e} + 6\mathbf{f}$$

$$\overrightarrow{EF} = -10.5\mathbf{e} - 21\mathbf{f}$$

(b) Find the ratio

length of  $DF$  : length of  $DE$

(3)